

Education

Ph.D. in Computing Science

UNIVERSITY OF ALBERTA

» Supervisors: Dale Schuurmans, Marlos C. Machado

2019 – 2026

M.Sc. in Statistics

UNIVERSITY OF ALBERTA

» Supervisors: Ivor Cribben, Rohana Karunamuni

2016 – 2018

Honours Bachelor in Mathematics

UNIVERSITY OF WATERLOO

» Major: Mathematical Economics

2012 – 2016

Papers & Preprints

Learning to Forget: Continual Learning with Adaptive Weight Decay

» A. A. Ramesh, **A. Lewandowski**, J. Schmidhuber
» Preprint

04/2026

Universal computation is intrinsic to language model decoding

» **A. Lewandowski**, M. C. Machado, D. Schuurmans
» Preprint

01/2026

The world is bigger: A computationally-embedded perspective on the big world hypothesis

» **A. Lewandowski**, A. Ramesh, E. Meyer, D. Schuurmans, M. C. Machado
» Neural Information Processing Systems (**NeurIPS spotlight**)

12/2025

A Unified Noise-Curvature View of Loss of Trainability

» G. S. Baveja, **A. Lewandowski**, M. Schmidt
» Workshop on Optimization for Machine Learning at NeurIPS

12/2025

Balancing expressivity and robustness: Constrained rational activations for reinforcement learning

» R. Surdej, M. Borkiewicz, **A. Lewandowski**, M. Ostaszewski, C. Lyle
» Conference on Lifelong Learning Agents (**CoLLAs oral**)

08/2025

Plastic learning with deep Fourier features

» **A. Lewandowski**, D. Schuurmans, M. C. Machado
» International Conference on Learning Representations (**ICLR**)

04/2025

Learning continually by spectral regularization

» **A. Lewandowski**, M. Borkiewicz, S. Kumar, A. György, D. Schuurmans, M. Ostaszewski, M. C. Machado
» International Conference on Learning Representations (**ICLR**)

04/2025

The need for a big world simulator: A scientific challenge for continual learning

» S. Kumar, H. J. Jeon, **A. Lewandowski**, B. Van Roy
» Finding The Frame Workshop at RLC (**RLC workshop oral**)

08/2024

Directions of curvature as an explanation for loss of plasticity

» **A. Lewandowski**, H. Tanaka, D. Schuurmans, M. C. Machado
» Preprint

11/2023

Reinforcement teaching

» C. Muslimani*, **A. Lewandowski***, D. Schuurmans, M. Taylor, J. Luo
» Transactions on Machine Learning Research (**TMLR**)

06/2023

ZORB: A derivative-free backpropagation algorithm for neural networks

» V. Ranganathan, **A. Lewandowski**
» Beyond Backpropagation Workshop at NeurIPS (**NeurIPS workshop oral**)

12/2020

*Equal contribution

Work Experience

Research Associate (Part-time)

NOAH'S ARK LAB, HUAWEI TECHNOLOGIES CANADA CO., LTD.2022 – 2023

» Supervisor: Jun Luo

» Developed a reinforcement learning framework for meta-learning to improve training speed of new agents

Research Associate (Full-time)

NOAH'S ARK LAB, HUAWEI TECHNOLOGIES CANADA CO., LTD.2021 – 2022

» Supervisor: Jun Luo

» Supervised two software developer interns who worked on a simulator for unprotected left turns for robust agents (ULTRA). Designed an evaluation methodology for generalization of reinforcement learning agents.

Teaching Assistant

UNIVERSITY OF ALBERTA2016 – 2018, 2019 – 2021

» Computing Science: Facilitated discussion groups and developed assignments for a new reinforcement learning course. Provided mentorship and guidance for student projects in foundations of information retrieval

» Mathematical & Statistical Sciences: Offered personalized support to students in all first and second-year classes at the Decima Robinson Support Centre. Conducted help sessions Statistics I/II, Regression Analysis and Time Series Analysis.

Honors & Awards

Alberta Innovates Graduate Student Scholarship

University of Alberta, Department of Computing Science2022–2024

President's Doctoral Prize of Distinction

University of Alberta, Department of Computing Science2021–2024

NSERC Postgraduate Scholarships – Doctoral

University of Alberta, Department of Computing Science2021–2024

Josephine Mitchell Scholarship

University of Alberta, Department of Mathematical and Statistical Sciences2017–2018

Profiling Alberta's Graduate Students Award

University of Alberta, Department of Mathematical and Statistical Sciences2017

Queen Elizabeth II Graduate Scholarship

University of Alberta, Department of Mathematical and Statistical Sciences2016

President's Scholarship

University of Waterloo2012

Community

Organizer

RL in Big Worlds at RLC 2026, RLC 2025 (Local Chair), Finding the Frame Workshop at RLC 2025, Openmind Research Institute Banff Retreat 2023-2024, ICLR SSBM Social 2021, NeurIPS SSBM Social 2020, ICLR Reinforcement Learning Social 2020

Reviewer

NeurIPS 2021-2026 (Top Reviewer: 2022, 2025), ICLR 2020-2026 (Outstanding Reviewer: 2021, 2022, 2026), ICML 2021-2024 (Outstanding Reviewer: 2022), RLC 2024-2026, CoLLAs 2025-2026

Other

Academic Director at Computing Science Graduate Student Association 2022–2023